# Session 3: CHALLENGES WITH CONVERGENCE

# CHALLENGES WITH CONVERGENCE

## Impact of Batch Size

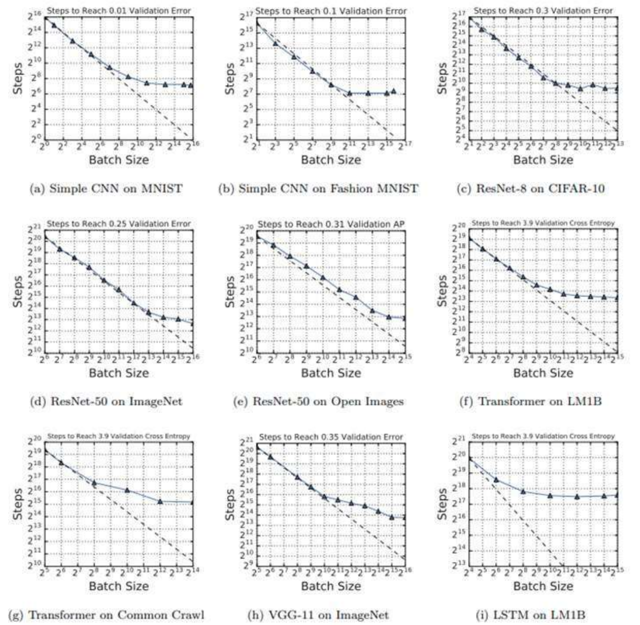


Figure 1: **The relationship between steps to result and batch size has the same characteristic form for all problems.** In all cases, as the batch size grows, there is an initial period of **perfect scaling** (indicated with a dashed line) where the steps needed to achieve the error goal halves for each doubling of the batch size. Then there is a region of **diminishing returns** that eventually leads to a region of **maximal data parallelism** where additional parallelism provides no benefit whatsoever. AP denotes average precision (see Appendix A).

Shallue, C. J., Lee, J., Antognini, J., Sohl-Dickstein, J., Frostig, R., & Dahl, G. E. (2018). Measuring the effects of data parallelism on neural network training. arXiv:1811.03600

# CHALLENGES WITH CONVERGENCE

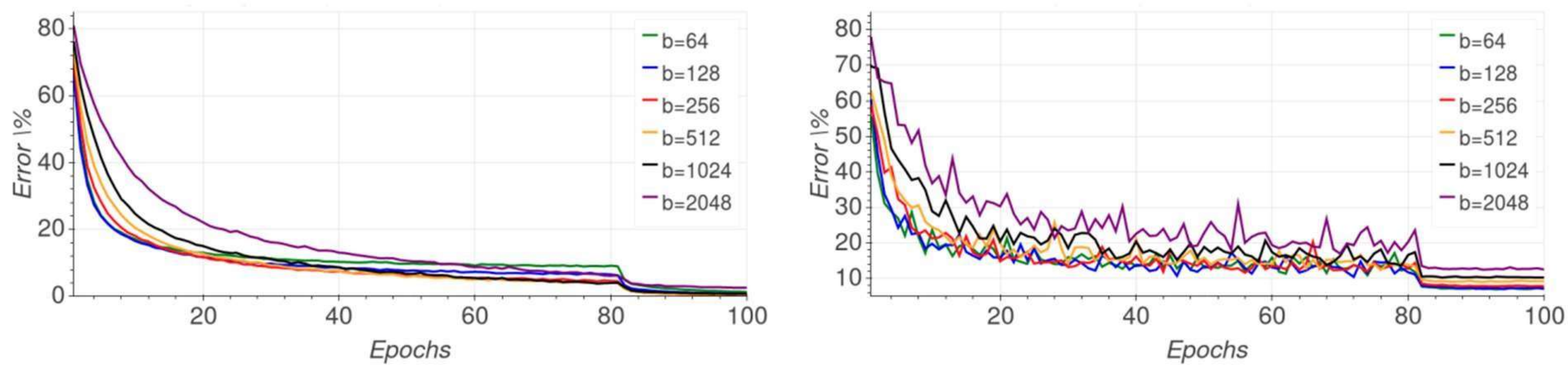
## Impact of Batch Size


"critical batch size" is the point at which compute efficiency drops below 50% optimal, at which point training speed is also typically 50% of optimal.

https://blog.openai.com/science-of-ai/

# CHALLENGES WITH CONVERGENCE

## Impact on Accuracy




(a) Training error

(b) Validation error

Figure 1: Impact of batch size on classification error

Hoffer, E., Hubara, I., & Soudry, D. (2017). Train longer, generalize better: closing the generalization gap in large batch training of neural networks. arXiv:1705.08741

# CHALLENGES WITH CONVERGENCE

## Techniques for faster convergence

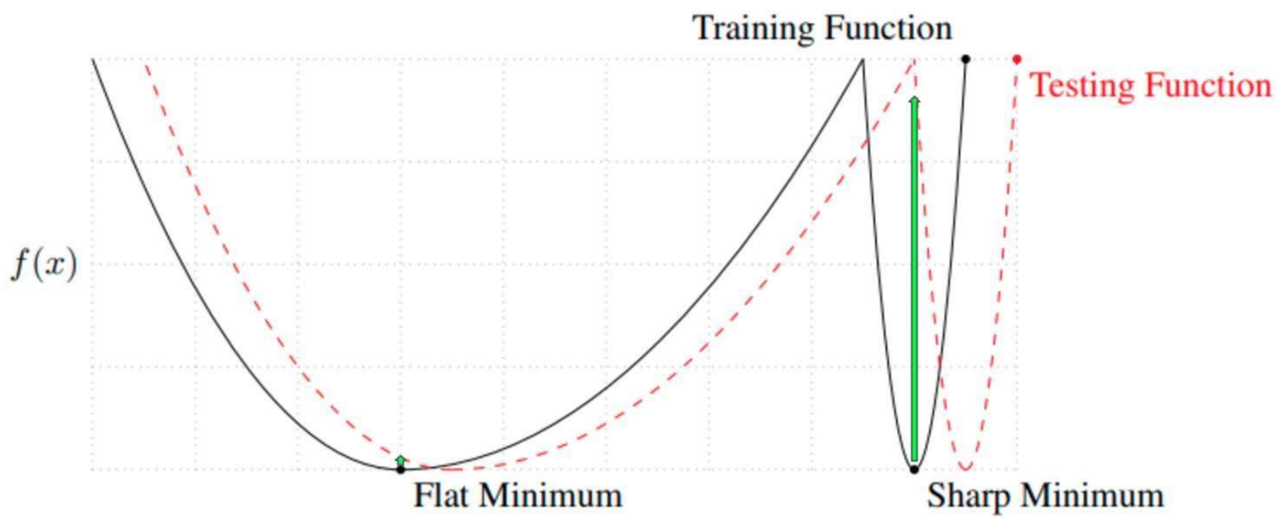



Figure 1: A Conceptual Sketch of Flat and Sharp Minima. The Y-axis indicates value of the loss function and the X-axis the variables (parameters)

Keskar, N. S., et al. (2016). On large-batch training for deep learning: Generalization gap and sharp minima. arXiv:1609.04836

# TECHNIQUES FOR FASTER CONVERGENCE

- Ghost batch Normalization
- Learning Rate Scaling
- Learning Rate Warmup
- Optimizers built for Exascale Deep learning

# TECHNIQUES FOR FASTER CONVERGENCE

## Ghost batch Normalization

```
X = BatchNormalization(...)(X)
```

**Ghost Batch Normalization.** Batch Normalization (BN) (Ioffe & Szegedy, 2015), is known to accelerate the training, increase the robustness of neural network to different initialization schemes and improve generalization. Nonetheless, since BN uses the batch statistics it is bounded to depend on the choosen batch size. We study this dependency and observe that by acquiring the statistics on small virtual ("ghost") batches instead of the real large batch we can reduce the generalization error. In our experiments we found out that it is important to use the full batch statistic as suggested by (Ioffe & Szegedy, 2015) for the inference phase. Full details are given in Algorithm 1. This modification by itself reduce the generalization error substantially.

Hoffer, E., Hubara, I., & Soudry, D. (2017). Train longer, generalize better: closing the generalization gap in large batch training of neural networks. arXiv:1705.08741

# TECHNIQUES FOR FASTER CONVERGENCE

## Learning rate scaling

### 2.1. Learning Rates for Large Minibatches

Our goal is to use large minibatches in place of small minibatches while *maintaining training and generalization accuracy*. This is of particular interest in distributed learning, because it can allow us to scale to multiple workers[2] using simple data parallelism without reducing the per-worker workload and without sacrificing model accuracy.

As we will show in comprehensive experiments, we found that the following learning rate scaling rule is surprisingly effective for a broad range of minibatch sizes:

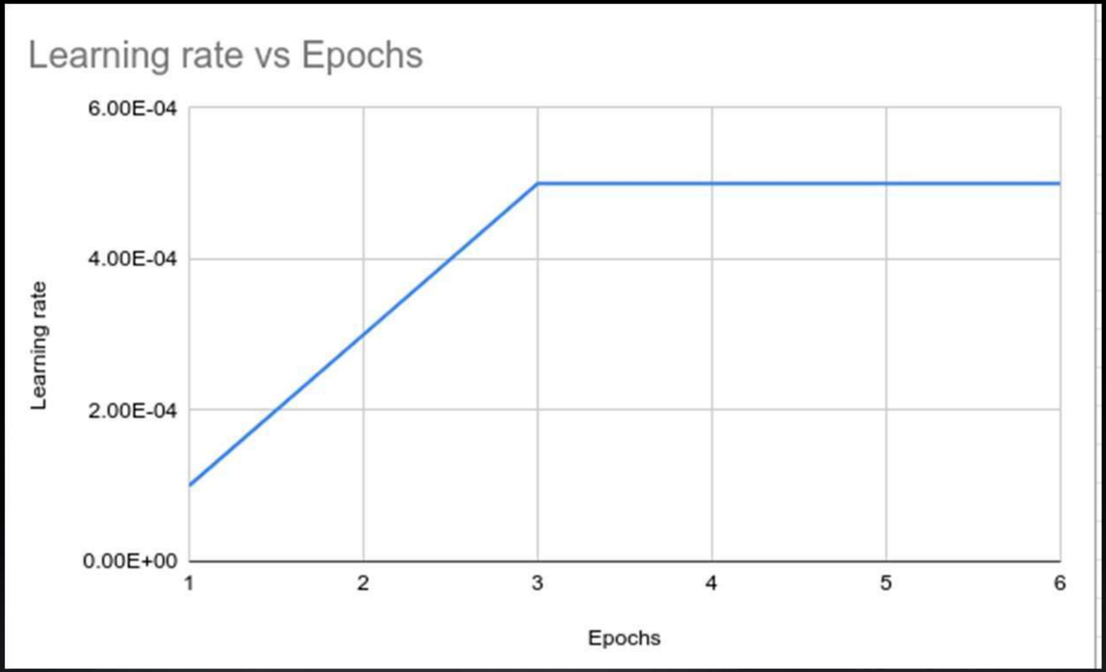
> ***Linear Scaling Rule:*** *When the minibatch size is multiplied by $k$, multiply the learning rate by $k$.*

All other hyper-parameters (weight decay, *etc*.) are kept unchanged. As we will show in §5, the *linear scaling rule* can help us to not only match the accuracy between using small and large minibatches, but equally importantly, to largely match their training curves, which enables rapid debugging and comparison of experiments prior to convergence.

Goyal, P., Dollár, P., Girshick, R., Noordhuis, P., Wesolowski, L., Kyrola, A., ... & He, K. (2017). Accurate, Large Minibatch SGD: Training ImageNet in 1 Hour. arXiv:1706.02677

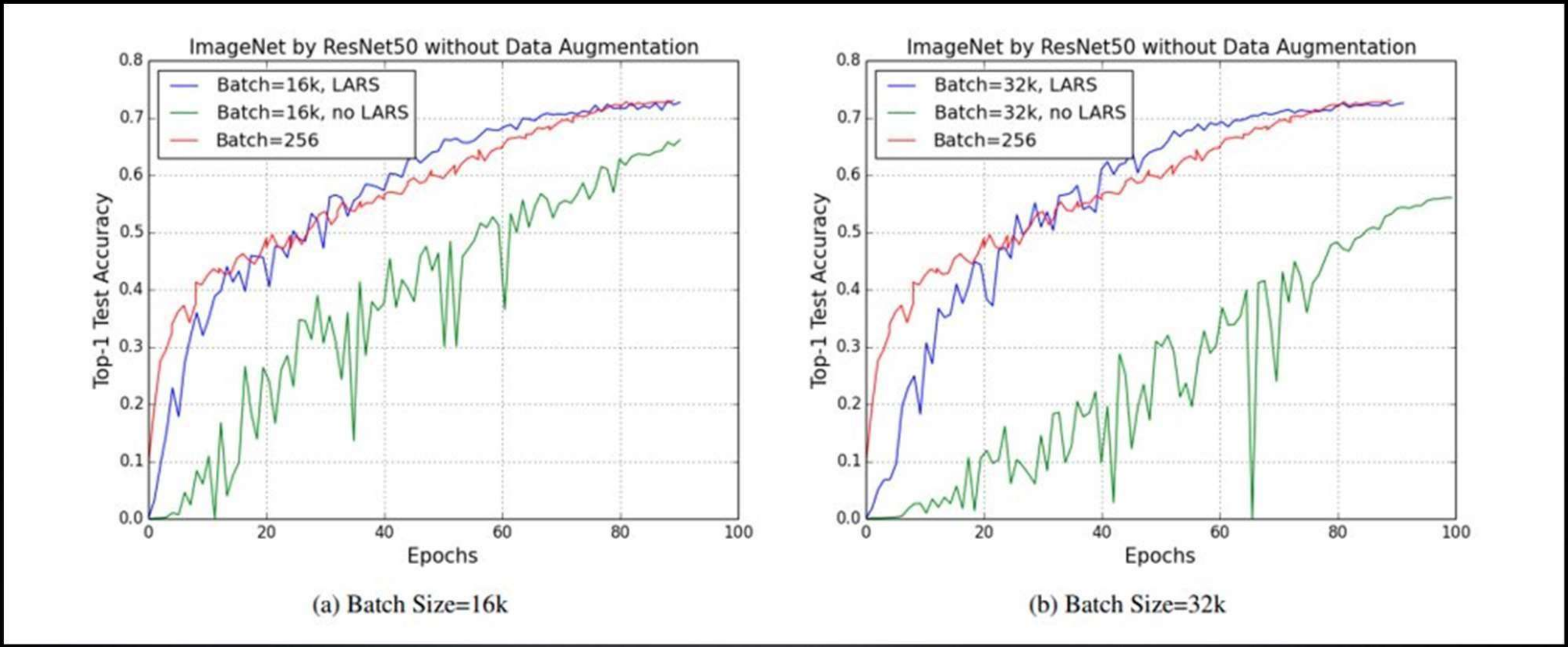
# TECHNIQUES FOR FASTER CONVERGENCE

## Learning rate warmup



# TECHNIQUES FOR FASTER CONVERGENCE

## Optimizers built for Exascale Deep Learning




You, Y., Zhang, Z., Hsieh, C., Demmel, J., & Keutzer, K. (2017). ImageNet training in minutes. arXiv: 1709.05011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# THANK YOU

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